

Feature Engineering

I. Categorical Predictors

K-level categorical $\rightarrow$ K-1 dummy variables

Insurance data price $\sim$ region

region = { NE, NW, SE, SW }

K=4

$$X = \begin{pmatrix} & \text{NW} & \text{SE} & \text{SW} \\ & 1 & 0 & 0 \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \\ & \vdots & \vdots & \vdots \end{pmatrix}$$

One-hot encoding

Northeast (NE) is called the reference category.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{NW} + \hat{\beta}_2 \text{SE} + \hat{\beta}_3 \text{SW}$$

NW 1 0 0

$\hat{\beta}_0$ = mean price for NE

Region	$\hat{Y}$	
NE	$\hat{\beta}_0$	$\hat{\beta}_0 = 50k$
NW	$\hat{\beta}_0 + \hat{\beta}_1$	$\hat{\beta}_1 = 5k$
SE	$\hat{\beta}_0 + \hat{\beta}_2$	
SW	$\hat{\beta}_0 + \hat{\beta}_3$	

$S_0, \hat{\beta}_j$ = difference in mean of Y between category j and reference category ($\hat{\beta}_0$)

- Ordinal categorical $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{educ.}$

Ex: education level = $\begin{matrix} \text{Some HS} & \text{HS} & \text{AA} & \text{BA} & \text{MA} & \text{PhD} \\ 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$

Caution: this implies there is a distance between categories.

II Interactions

Does effect of X_1 on Y depend on value of X_2 ?

Ex: does effect of age on price depend on region?

Categorical + Continuous

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{Age} + \hat{\beta}_2 \text{Smoker} + \hat{\beta}_3 \text{Age} \times \text{Smoker}$$

interaction

Smoker

No = 0

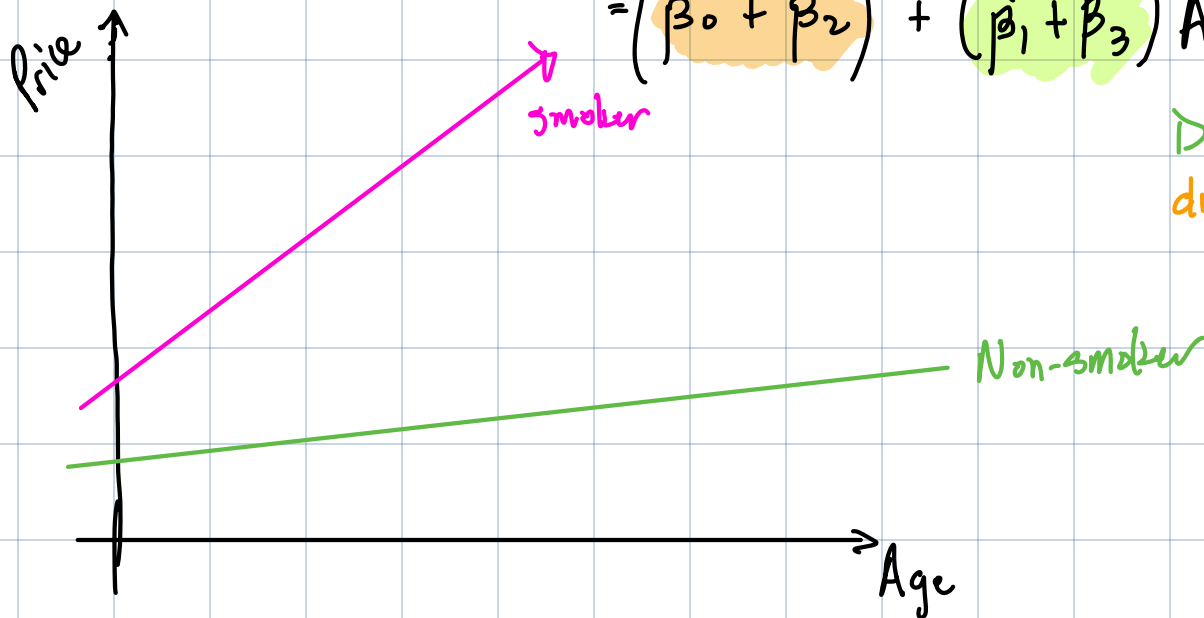
Yes = 1

When Smoker = No = 0

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{Age}$$

Otherwise

$$\begin{aligned} \hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 \text{Age} + \hat{\beta}_2 + \hat{\beta}_3 \text{Age} \\ &= (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) \text{Age} \end{aligned}$$



Different slopes,
different intercepts

Continuous + Continuous

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{Age} + \hat{\beta}_2 \text{Income} + \hat{\beta}_3 \text{Age} \times \text{Income}$$

$$(\hat{\beta}_1 + \hat{\beta}_3 \text{Income}) \text{Age}$$

function of income

Categorical + Categorical

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{Smoker} + \hat{\beta}_2 \text{Alcohol} + \hat{\beta}_3 \text{Smoker} \times \text{Alcohol}$$

		Smoker	
		No (0)	Yes (1)
Alc	No (0)	$\hat{\beta}_0$	$\hat{\beta}_0 + \hat{\beta}_1$
	Yes (1)	$\hat{\beta}_0 + \hat{\beta}_2$	$\hat{\beta}_0 + \hat{\beta}_2 + \underbrace{\hat{\beta}_1 + \hat{\beta}_3}_{\text{additional effect of smoking}}$

Note: If $X_1 \times X_2$ is in model, include main effects X_1 and X_2

$$X_1 \times X_2 \times X_3$$

$$X_1, X_2, X_3$$

$$X_1 X_2, X_1 X_3, X_2 X_3$$

$$X_1 X_2 X_3$$

III

Centering + Standardizing

mean income = 50

① Centering: $X_i - \bar{X}_i$

Income
50k
60k
62k
i

Income_c
0k
10k
12k
i

do this for interactions

$\hat{\beta}_j$ = "Effect of X_j at average value of X_{j+1} "

(2) Standardizing

$$Z = \frac{X_j - \bar{X}_j}{sd(X_j)}$$

put all X 's on same scale
of std devs from the mean

Interpretation: "for each 1-sd increase, y changes by β ."